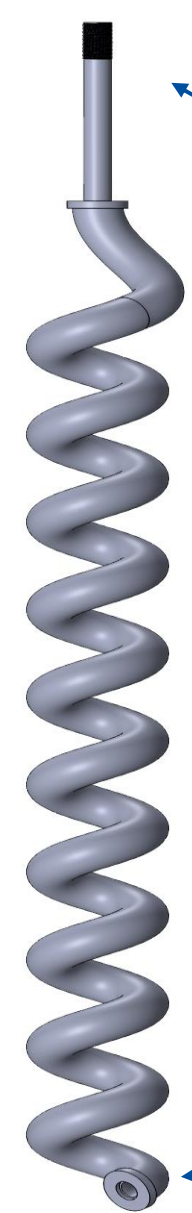


Cooling mechanism –

The helical rod

- A helical tube composed of an outer insulating layer, cooling gel and stainless steel inside layer.
- Maximises thermal capacity and convection, cooling only its inside flow.
- Rods can be purchased separately to switch when needed in operation.



Screwed into device here,
Fit into bottle with cork

Shape
maximises
fluid contact
time

Design Objective

- Develop the design for a device with a **built-in chiller** which can pump beverages such as juice.
- Aimed to an approximate production run of **10,000** products.
- Designed for easy to use and ergonomic for **TROPICAL** countries beverage markets.

Drive train selection

Motor calculation:

Chosen Motor	Nominal Voltage	No Load Speed	No Load Current	Stall Torque	Stall Current
RF-500TB-14415	5	3600 rpm	0.0026	6.86 mNm	0.54 A

Battery Selection: Varta V150H 4.8V Button rechargeable battery

Nominal voltage: **5V** Capacity: **150 mAh**

Gear Selection:

Reduction ratio = $\frac{3600}{236} = 15.3$ (single stage)

Reduction ratio = $\sqrt{15.3} = 3.9$ (2 stages)

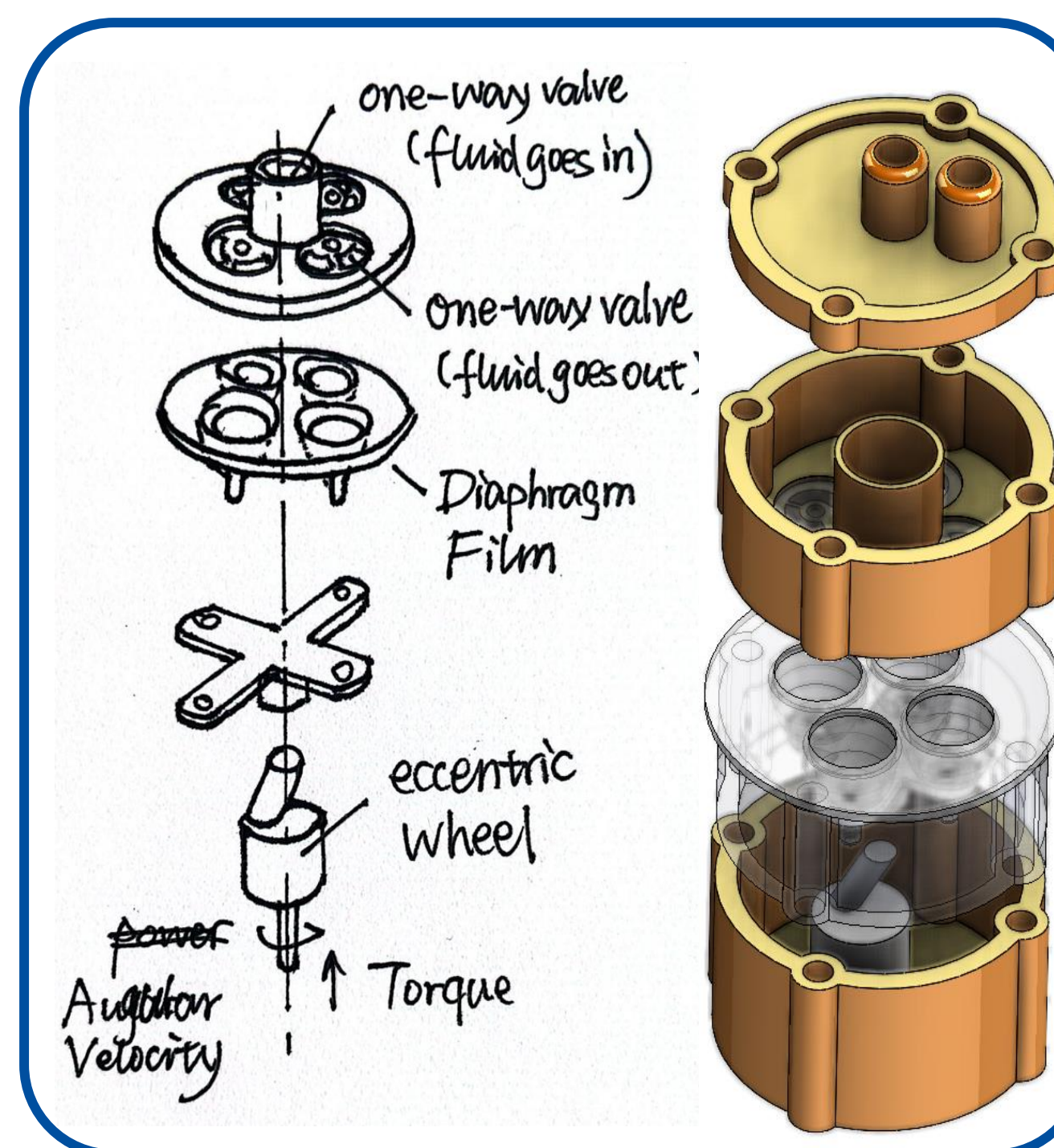
Gear Number	Teeth	PCD [mm]	ø Boss [mm]	ø Bore [mm]
ZG0.4-10	10	4.4	5.2	1.5
ZG0.4-39	27	15.6	11	3

Material Selection

Part	Material
Cooling gel	Propylene Glycol
Gear	Polyoxymethylene (POM)
Shaft	Aluminum alloys
Cooling rod inner wall	Stainless steel
Cooling rod outer wall	High-density polyethylene
Cork	Polyisoprene rubber(IIR)
Pump Case	Polypropylene
Pump diaphragm	Natural rubber
Pump rod	High-density polyethylene

Final
Concept
Render

Fits the
average
750ml
bottle



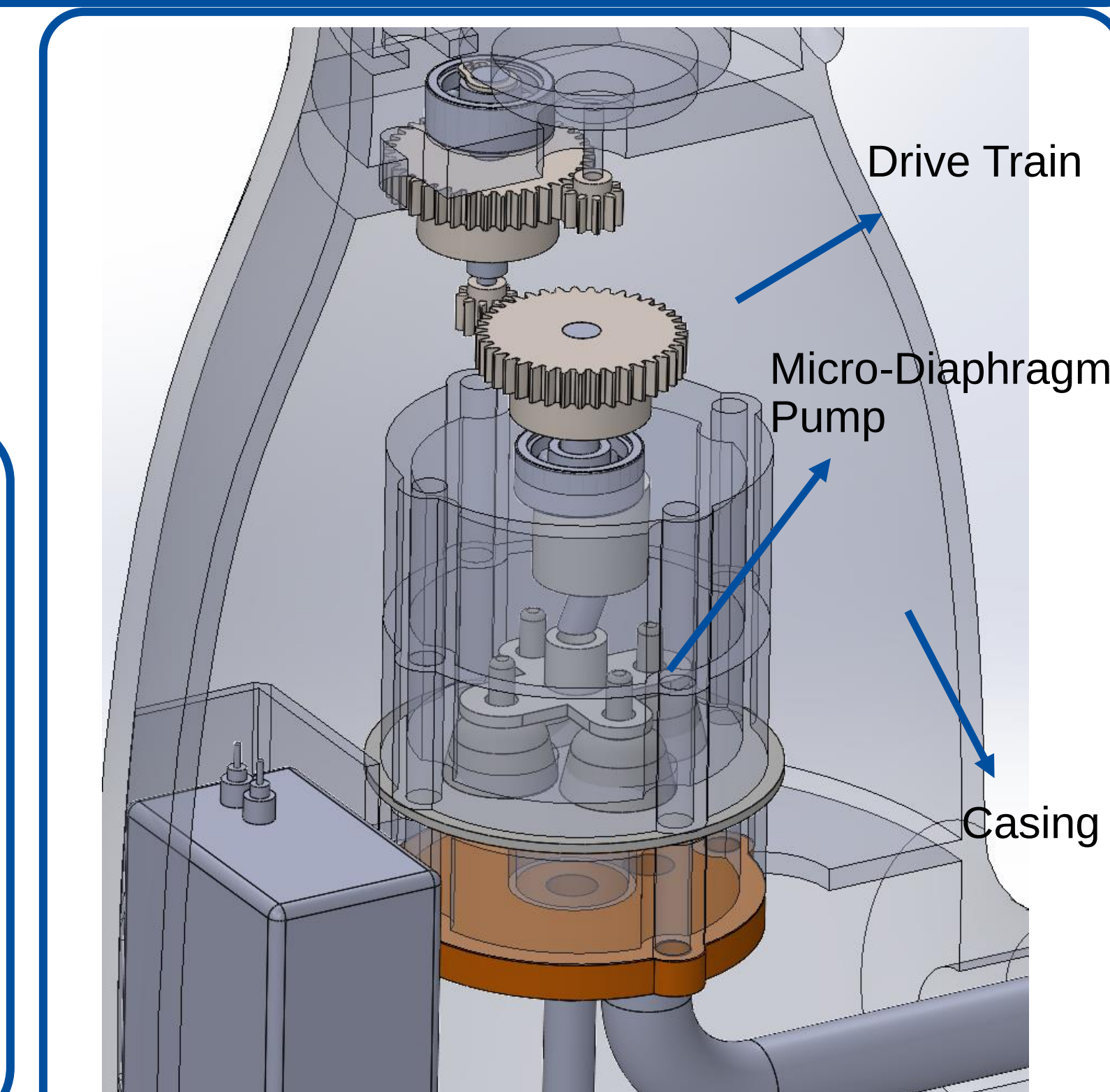
Function of Micro-Diaphragm Pump

Why choose a Micro-Diaphragm Pump?

- Pump generates large **pressure difference**
 - Can handle juices with different consistencies
 - Good control of flow rate
- Easy to manufacture (sustainability)
 - Use of additive manufacturing ∴ less waste

Manufacturing conditions:

- 3D printed and silicone diaphragm through injection molding



References: [Forced Convection h](#); [Functioning of pump](#)